

Anup Regmi

Senior Data Engineer

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Senior Data Engineer with 5+ years of experience designing and implementing scalable **Azure-based data platforms, lakehouse architectures, and enterprise ETL/ELT pipelines** across healthcare, retail, and financial services domains. Strong expertise in **Azure Databricks, Delta Lake, Snowflake, dbt, Azure Synapse Analytics, and Apache Spark**, building high-performance batch and real-time data pipelines processing multi-terabyte datasets. Proven track record of delivering **HIPAA-compliant, analytics-ready data models** using Medallion Architecture and dimensional modeling techniques. Experienced in **CI/CD automation, Infrastructure as Code (Terraform), data governance, and performance optimization**, enabling reliable, secure, and cost-efficient cloud data ecosystems.

TECHNICAL SKILLS

- **Cloud & Data Platforms:**
Microsoft Azure, Azure Data Factory (ADF), Azure Databricks, Azure Synapse Analytics, Azure Data Lake Storage Gen2 (ADLS), Azure Event Hubs, Azure Functions, Azure Logic Apps, Snowflake
- **Data Engineering & Processing:**
ETL / ELT Development, Apache Spark (PySpark), Delta Lake, Medallion Architecture (Bronze/Silver/Gold), Batch & Real-Time Data Pipelines, Data Lakehouse Architecture
- **Streaming & Event Processing:**
Spark Structured Streaming, Azure Event Hubs, Apache Kafka, Near Real-Time Processing
- **Data Warehousing & Modeling:**
Azure Synapse (Dedicated & Serverless SQL Pools), Snowflake Warehousing, Dimensional Modeling (Star & Snowflake Schema), Data Modeling, dbt Transformations
- **Programming & Query Languages:**
Python, SQL, PySpark, Java (MapReduce)
- **Data Governance & Security:**
HIPAA Compliance, Azure RBAC, Azure Active Directory (AAD), Managed Identities, Azure Key Vault, Data Masking, Encryption (At Rest & In Transit), Data Lineage
- **DevOps & Infrastructure:**
Terraform (Infrastructure as Code), Azure DevOps, Git, CI/CD Pipelines, Docker
- **Monitoring & Operations:**
Azure Monitor, Azure Log Analytics, Performance Optimization, Pipeline Alerting, SLA Management

PROFESSIONAL EXPERIENCE

Senior Data Engineer
Change Healthcare
Nashville, TN

June 2023- Present

- Architected a HIPAA-compliant Azure lakehouse platform using **Azure Databricks, Delta Lake, ADLS Gen2, and Snowflake**, supporting multi-terabyte healthcare claims and provider datasets.
- Designed scalable **ETL/ELT pipelines** in Azure Data Factory and Databricks to ingest structured and semi-structured healthcare data from EDI, APIs, and relational systems.
- Engineered high-performance **PySpark transformations** in Databricks, optimizing partitioning and Delta Lake file compaction to reduce query latency by 40%.
- Implemented Medallion Architecture (Bronze/Silver/Gold) to standardize ingestion, curation, and analytics-ready data modeling.
- Developed analytics-ready dimensional models in **Azure Synapse and Snowflake using dbt**, ensuring consistent metric definitions across finance and compliance teams.
- Enabled near real-time fraud detection signals using **Azure Event Hubs and Spark Structured Streaming**, processing thousands of events per second.

- Strengthened data governance by enforcing **Azure RBAC, Managed Identities, Key Vault encryption, and audit logging** aligned with HIPAA compliance standards.
- Established automated **data quality validation frameworks** using dbt tests and Spark-based reconciliation checks, improving reporting accuracy by 25%.
- Automated CI/CD workflows using **Azure DevOps, Git, and Terraform**, enabling controlled multi-environment pipeline deployments.
- Provisioned and standardized Azure infrastructure (Databricks workspaces, Synapse pools, Snowflake integrations) using Infrastructure as Code.
- Mentored junior engineers on Spark performance tuning, Delta Lake optimization, and Snowflake warehouse cost management.
- Reduced end-to-end pipeline execution time by 35% through query optimization, broadcast joins, and cluster auto-scaling configurations.
- Supported ingestion and secure processing of 5+ TB daily healthcare datasets with strict access controls, retention policies, and compliance auditing.

Key Technologies: Azure Databricks, Delta Lake, Azure Data Factory (ADF), Azure Synapse Analytics (Dedicated & Serverless SQL Pools), Azure Data Lake Storage Gen2 (ADLS), Snowflake, dbt, Apache Spark (PySpark), Azure Event Hubs, Spark Structured Streaming, Medallion Architecture (Bronze/Silver/Gold), Python, SQL, Terraform, Azure DevOps, Git, Azure Key Vault, Azure Active Directory (AAD), RBAC, Azure Monitor, HIPAA Compliance

Data Engineer
Home Depot, Washington, DC

Jan 2022 - May 2023

- Built enterprise-grade Azure and Snowflake-based data pipelines supporting retail sales, merchandising, and supply chain analytics across 100+ business users.
- Designed robust **ADF ingestion workflows** integrating POS systems, vendor feeds, APIs, and transactional databases into ADLS Gen2.
- Developed scalable Spark-based ETL pipelines in **Azure Databricks**, transforming raw sales data into curated analytical datasets.
- Centralized structured and semi-structured retail data into a governed **Azure Data Lake + Snowflake architecture**, enabling unified reporting.
- Modeled star schemas and fact-dimension tables in **Azure Synapse and Snowflake using dbt**, improving dashboard performance by 30%.
- Implemented near-real-time inventory updates using **Event Hubs and Structured Streaming**, reducing stock reconciliation delays.
- Standardized ELT transformation logic using **dbt modular models**, improving maintainability and reducing code duplication by 25%.
- Partnered with business stakeholders to define data contracts, KPIs, and metric governance for consistent financial reporting.
- Applied SQL optimization and clustering strategies in Snowflake warehouses to reduce compute cost by 20%.
- Strengthened data reliability by implementing automated data validation checks and reconciliation pipelines.
- Automated release management using Git-based CI/CD pipelines, reducing manual deployment effort by 40%.
- Engineered Snowflake data masking, RBAC policies, and secure data sharing for compliance-sensitive datasets.
- Built cross-cloud integrations using Terraform to provision Azure, Databricks, Snowflake, and AWS S3/EMR environments where required.
- Enhanced pipeline fault tolerance using retry logic, monitoring alerts, and SLA-driven workflow orchestration.
- Improved overall data freshness and availability by 30% through optimized Spark execution plans and workload isolation strategies

Key Technologies: Azure Data Factory (ADF), Azure Databricks, Azure Data Lake Storage Gen2 (ADLS), Azure Synapse Analytics, Snowflake, dbt, Apache Spark (PySpark), Azure Event Hubs, Spark Structured Streaming, SQL, Python, Terraform, Git, Azure DevOps, Snowflake Warehousing & Clustering, Data Modeling (Star Schema), Data Quality Validation, CI/CD Automation

- Designed and developed scalable Big Data pipelines using **Hadoop, Spark, and MapReduce**, processing multi-terabyte financial transaction datasets for risk and compliance analytics.
- Engineered batch and streaming **ETL workflows** using Apache Sqoop, Flume, and Kafka to ingest structured and semi-structured data into HDFS.
- Optimized Hive and Spark SQL queries using partitioning, bucketing, and execution plan tuning, reducing reporting latency by 25%.
- Built distributed data transformation pipelines using Apache Pig and Spark to standardize high-volume transactional records.
- Implemented real-time data ingestion pipelines using **Kafka**, enabling near real-time fraud detection and anomaly monitoring.
- Developed custom MapReduce jobs in Java for complex aggregations and financial reconciliation processes.
- Automated workflow scheduling and dependency management using Apache Oozie, ensuring SLA-driven batch processing.
- Modeled and managed semi-structured financial datasets in **HBase**, improving low-latency data retrieval for reporting systems.
- Improved HDFS storage efficiency through partitioning strategies and compression techniques, reducing storage footprint and improving job performance.
- Monitored and tuned Hadoop cluster performance using Ambari and Ganglia, increasing system reliability and resource utilization.
- Collaborated with analytics and data science teams to design reusable data models and curated datasets for downstream BI consumption.
- Contributed to modernization initiatives by supporting Spark-based transformations and evaluating migration strategies toward cloud-based analytics platforms.

Key Technologies: Hadoop, Apache Spark, MapReduce (Java), Apache Hive, Apache Pig, Apache Kafka, Apache Sqoop, Apache Flume, Apache Oozie, HBase, HDFS, Oracle, MySQL, JDBC, Ambari, Ganglia, Spark SQL, ETL Development, Data Warehousing Concepts, Performance Tuning

EDUCATION

Bachelors in Electrical and Computer Engineering, Texas A&M University, 400 Bizzell St, College Station, TX

CERTIFICATIONS

- *Microsoft Certified: Azure Data Engineer Associate (DP-203)*
- *Microsoft Certified: Azure Solutions Architect Expert (AZ-305)*